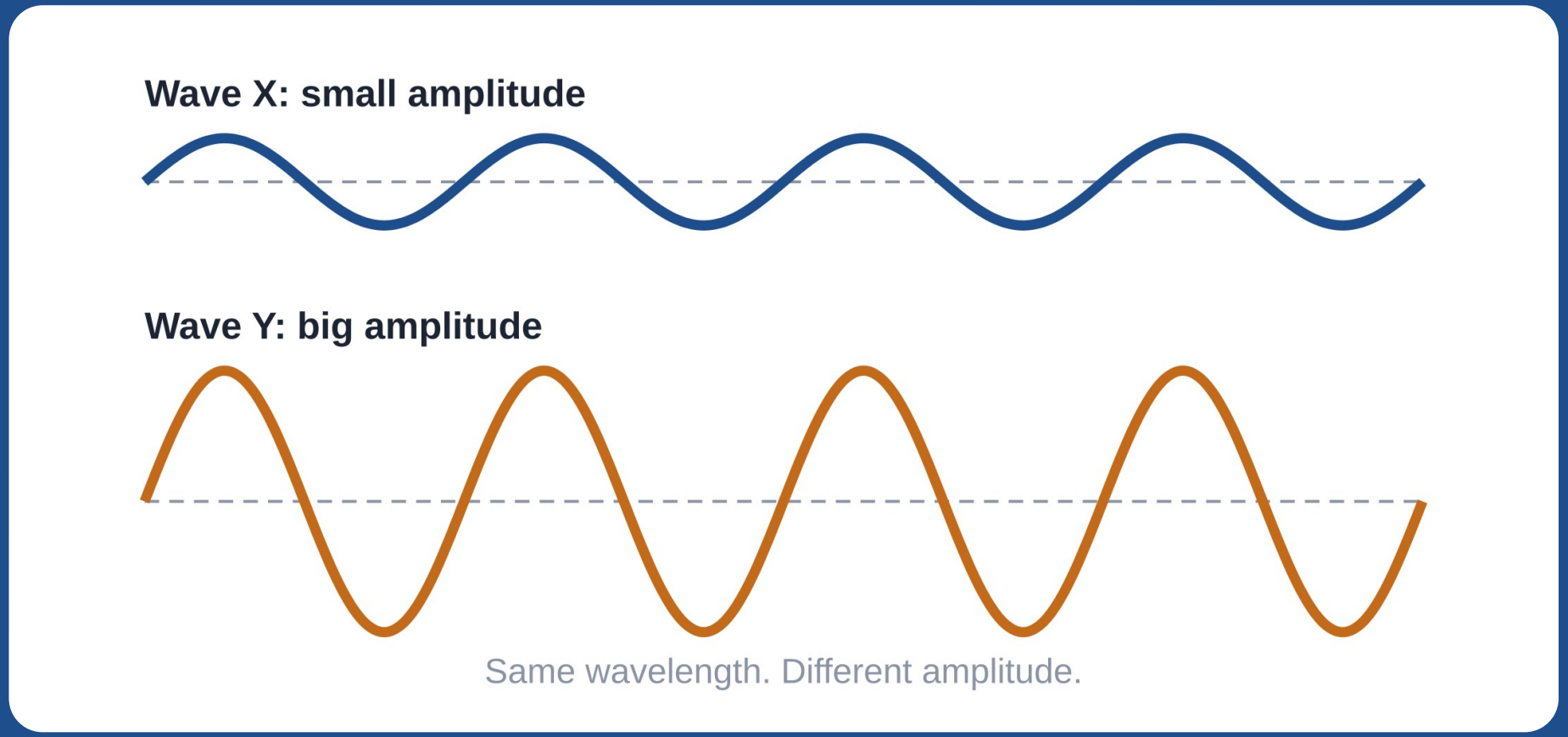
4.02 Amplitude

How does amplitude affect the energy of a wave?



Today's Goals

I can measure amplitude from the rest position to a crest or trough.

I can explain that a bigger amplitude means more energy.

I can give real examples: louder sounds, a rope shaken harder, stronger earthquakes.

I can show that changing amplitude does not change wavelength or frequency.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Waves can be described by their speed, wavelength, amplitude, and frequency.
- Vibrations cause wave-like disturbances that carry energy.
- **The energy of a mechanical wave increases with amplitude.**

Inquiry skills (Grades 5–8): "Think critically and logically to connect evidence and explanations" and "Analyze and interpret data" (today's argument and graphs).

In this lesson

- Integer warm-up
- Vocab check-in
- Measuring amplitude
- Amplitude and energy
- What amplitude does NOT change
- Who is right? argument
- Graph polls

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

CASE #3 • WEDNESDAY

Is Crest B past the pier post?

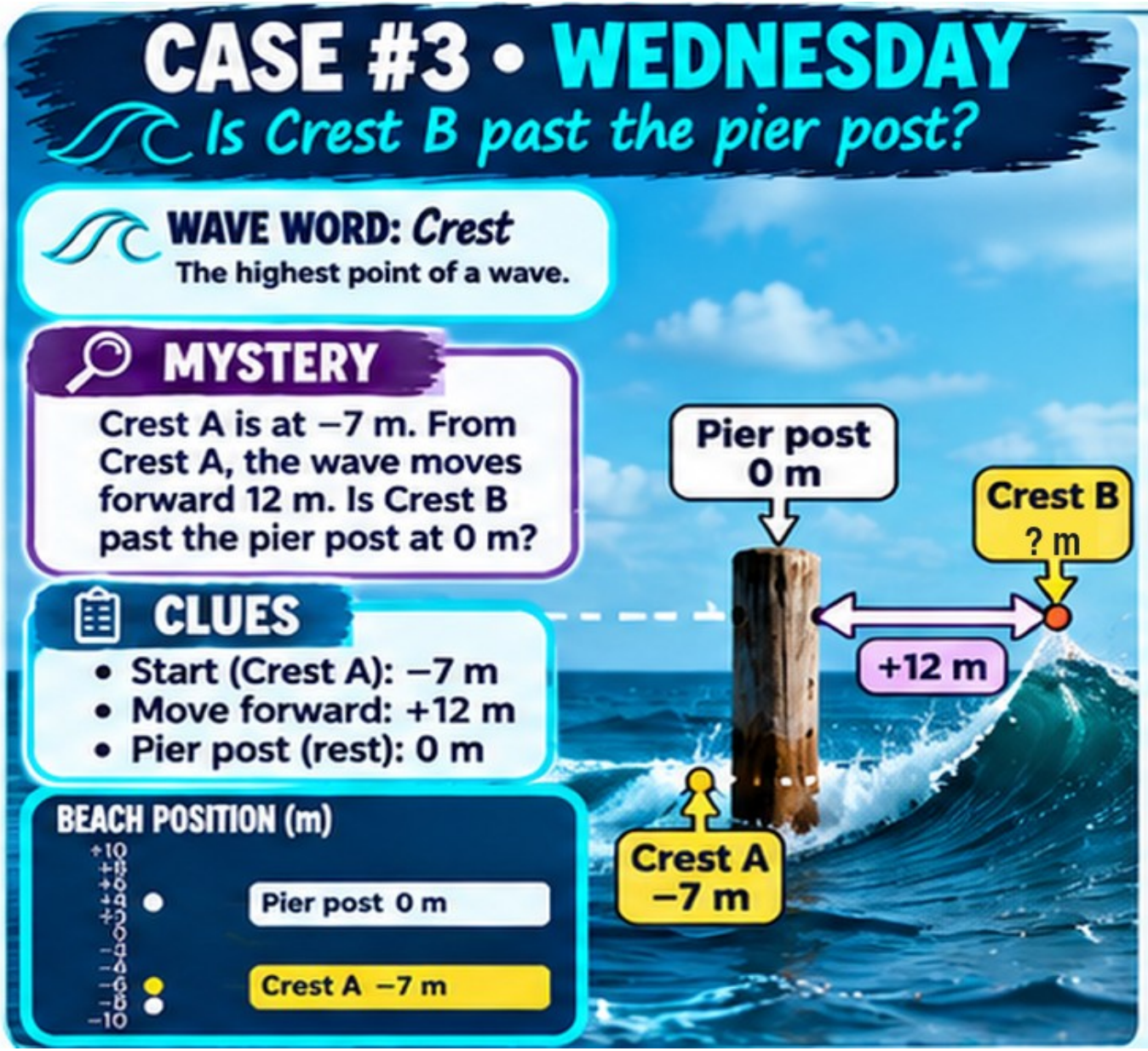
WAVE WORD: *Crest*
The highest point of a wave.

MYSTERY

Crest A is at −7 m. From Crest A, the wave moves forward 12 m. Is Crest B past the pier post at 0 m?

CLUES

- Start (Crest A): −7 m
- Move forward: +12 m
- Pier post (rest): 0 m



CLAIM

Answer the mystery question.

Is Crest B past the pier post?

☐ YES ☐ NO

EVIDENCE

Show your work!

Solve:

−7 + 12 = _____

REASONING

Explain the rule.

Finish the sentence:

"12 has the larger absolute value, so the answer is..."

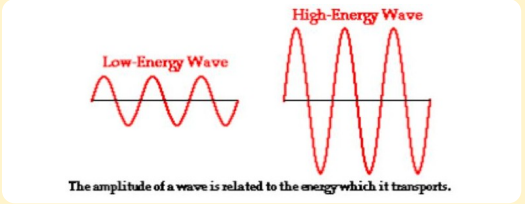
Vocab Check-In

From your Unit 4 vocabulary pretest. Today we focus on the starred words.



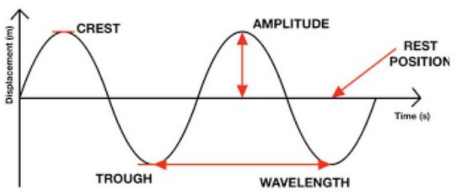
wave

a disturbance that carries energy from one place to another



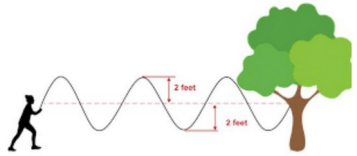
energy ★

the ability to do work or cause change



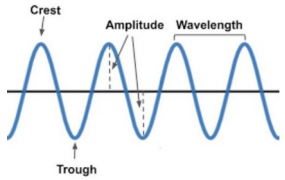
amplitude ★

how far a wave moves from its rest position



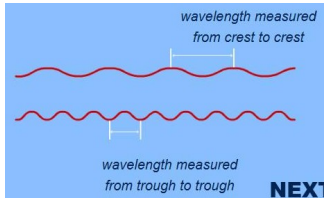
medium

the material a wave travels through (rope, string, air)



crest / trough ★

the highest point / the lowest point of a wave



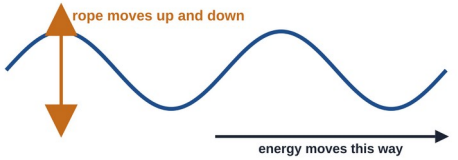
wavelength

the distance from one crest to the next crest



frequency

how many waves pass a point in a given time

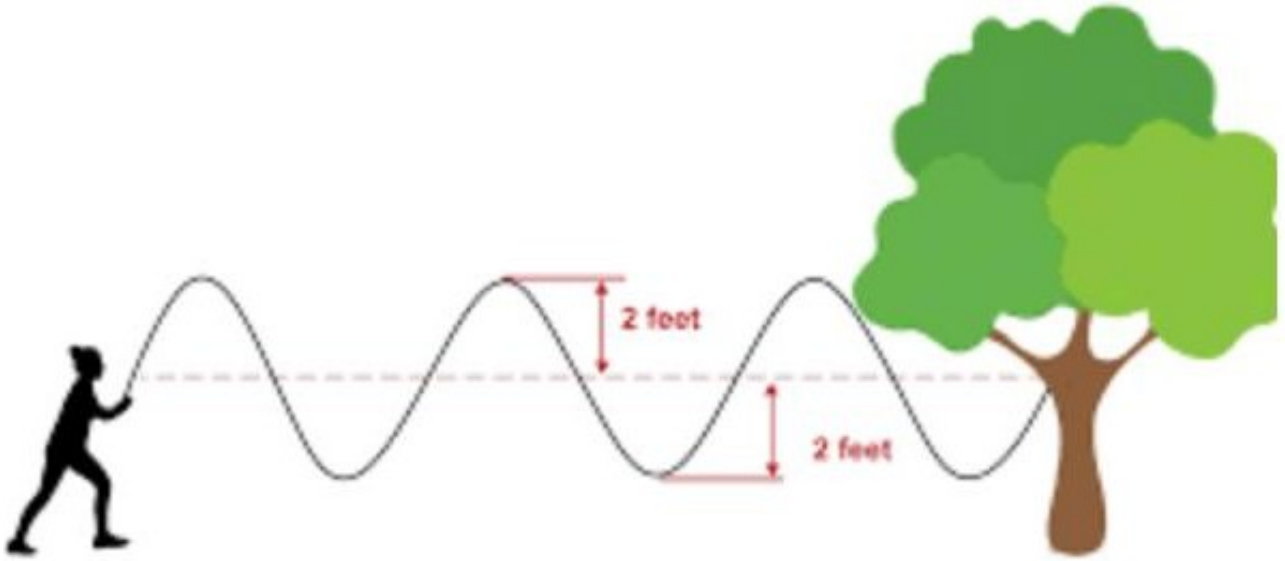


transverse wave

the medium moves up and down, across the way the wave travels

Classkick: 4.02 Part 1 · Q1–Q2

Measuring Amplitude



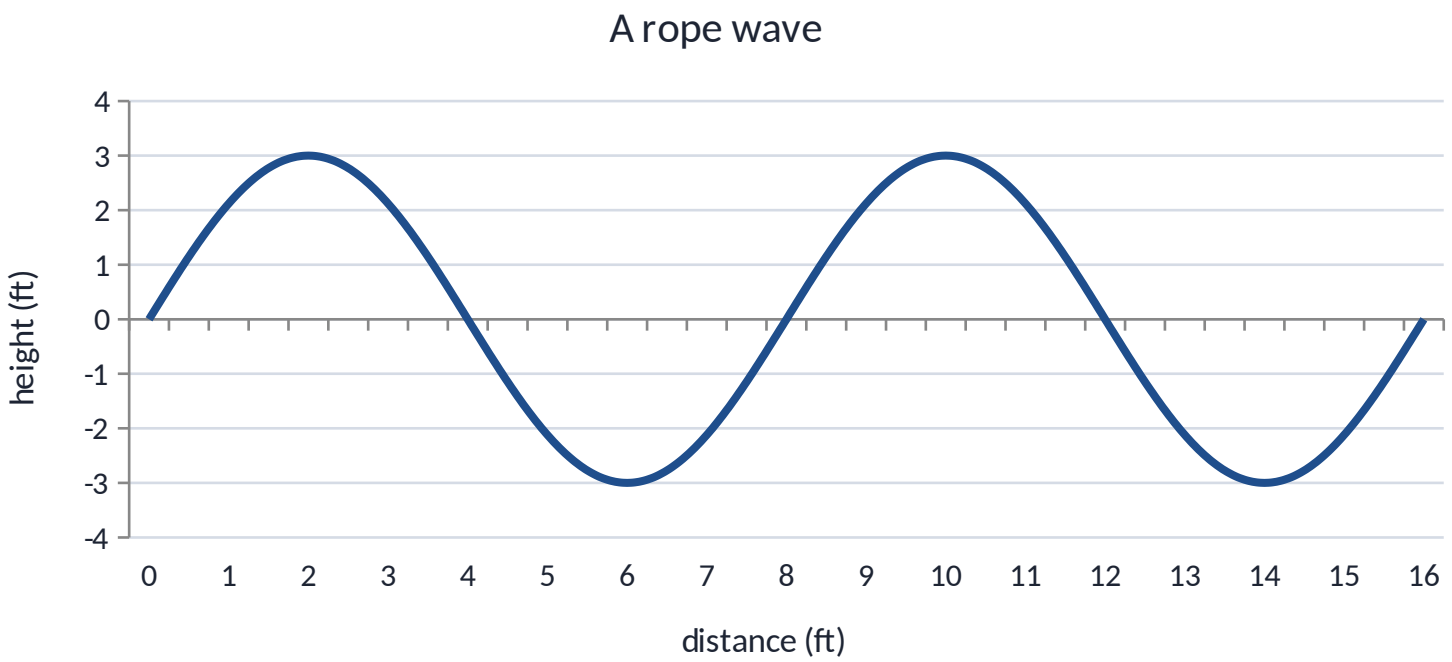
Amplitude = how far the wave moves from its **rest position** to a crest or a trough.

Up to the crest: 2 ft. Down to the trough: 2 ft. Same both ways!

Watch out! Crest to trough is NOT the amplitude. This rope wave is 4 ft from trough to crest, so its amplitude is only 2 ft.

Classkick: 4.02 Part 1 · Q2

Graph Poll



Read the graph.

What is the amplitude of this wave?

Hint: start at the rest position (0).

Single choice poll

A. 3 ft

Single choice poll

B. 6 ft

Single choice poll

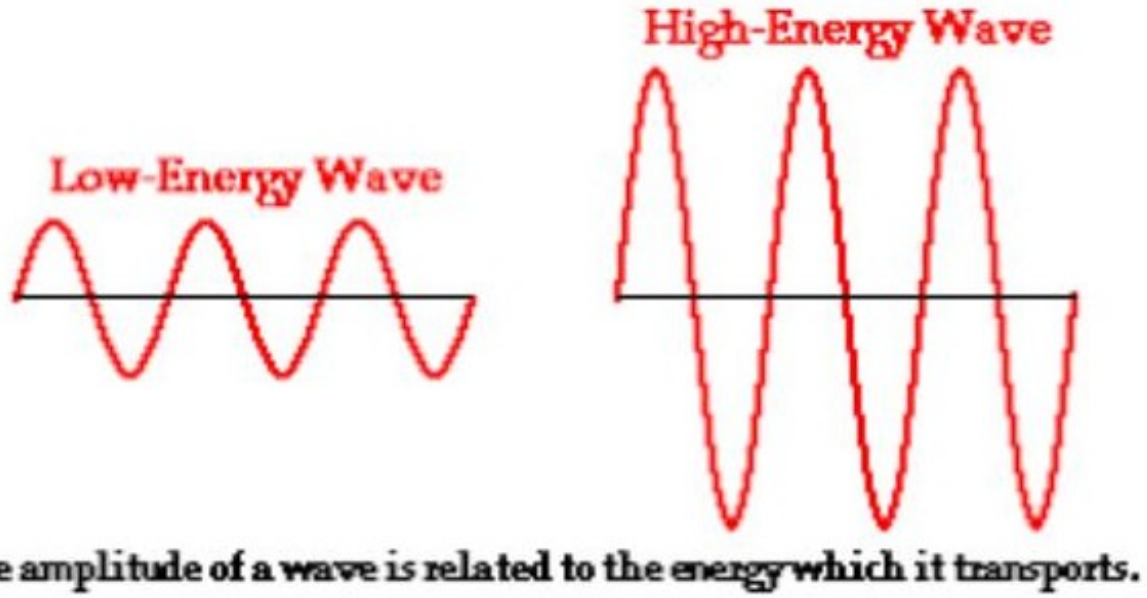
C. 8 ft

Single choice poll

D. 16 ft

Classkick: 4.02 Part 1 · Q3–Q6

Bigger Amplitude = More Energy



Which kind of wave carries MORE energy? (Q3)

PBS Amplitude Demonstration

Click "amplitude," then move the blue slider.

Pluck a guitar harder

bigger amplitude, louder sound

Turn up the volume

bigger amplitude, more energy

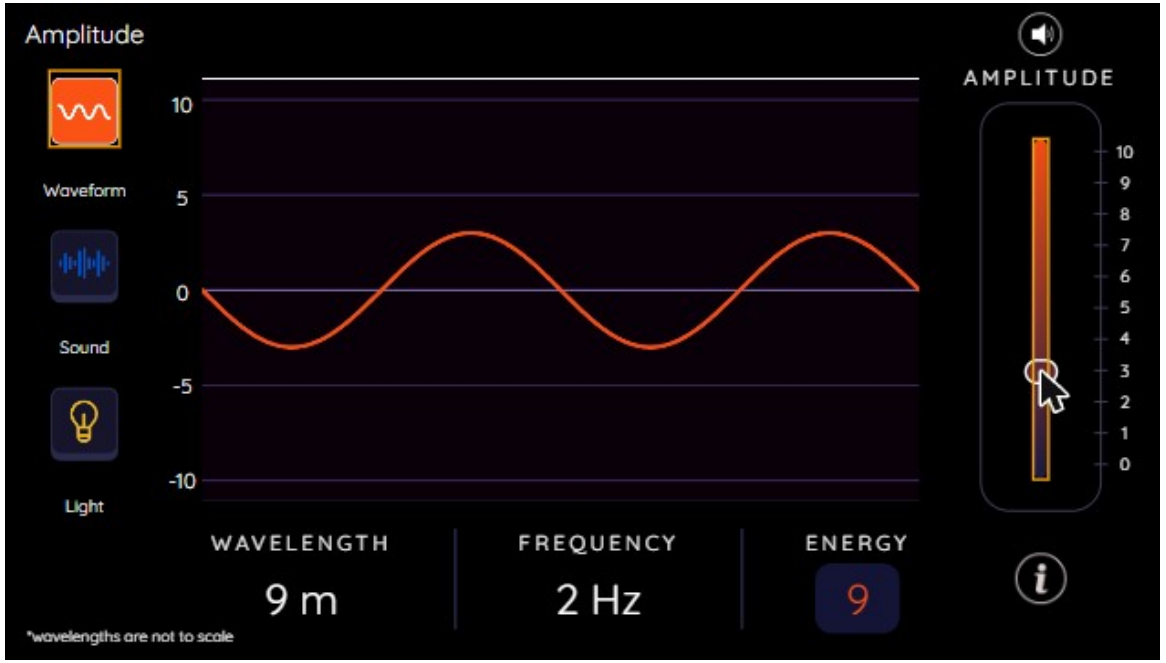
Shake a rope harder

taller waves, more energy

Stronger earthquakes

larger amplitude seismic waves

PBS Amplitude Demonstration



Low amplitude: energy 9



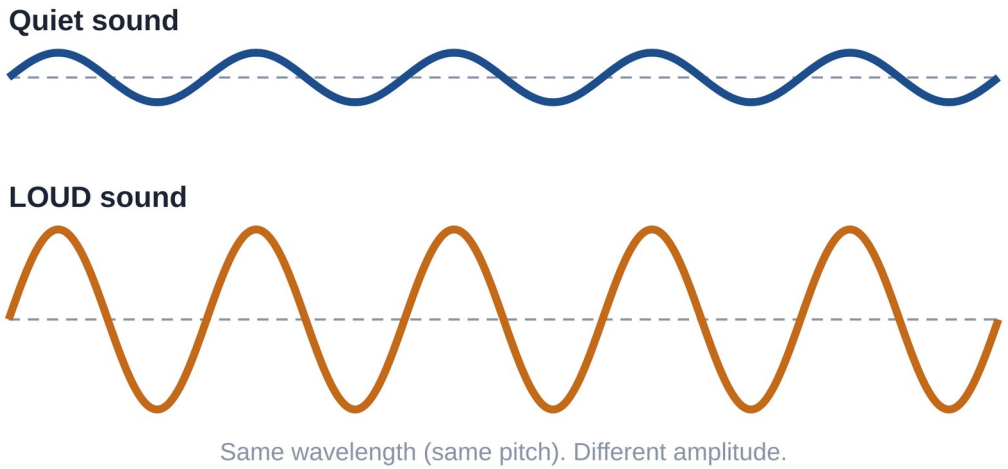
High amplitude: energy 64

Try it: click Waveform, then drag the Amplitude slider. Watch the ENERGY number. Did the wavelength or frequency change?

▶️ **Open the PBS interactive**

Engageli Poll

Classkick: 4.02 Part 1 · Q6



Same wavelength (same pitch). Different amplitude.

You turn UP the volume on a speaker. What happens to the sound waves?

Single choice poll

A. They travel faster

Single choice poll

B. Bigger amplitude, more energy

Single choice poll

C. Higher pitch

Single choice poll

D. Longer wavelength

Explore Amplitude in PhET

Water screen (Q7)

Turn on the water. Move the Amplitude slider low, then high.

Look for: taller waves, drops hit harder, more energy.

Sound screen (Q8)

Turn the sound ON. Move the Amplitude slider up.

Look for: louder sound, speaker moves farther. Does the pitch change?

▶️ Open PhET Waves Intro

What Amplitude Does NOT Change

When amplitude goes up…

Height of the wave	changes
Energy	changes
Wavelength (crest to crest)	stays the same
Number of waves (frequency)	stays the same

▶️ Simple Wave Simulator (Q9)

Sum it up (Q10)

Amplitude is how far the medium moves from its _____. A wave with a bigger amplitude carries more _____. A sound with a bigger amplitude is _____. Changing the amplitude does not change the _____.

Who Is Right?

Maya shakes a jump rope harder than Leo. Only her waves knock over a cup at the other end. Why?

Ava says:

"Maya's waves travel faster, so they hit the cup harder."

Ben says:

"Maya's waves carry the rope itself down to the cup."

Cam says:

"Maya's waves have a bigger amplitude, so they carry more energy to the cup."

Dee says:

"Maya's waves have a longer wavelength, so they reach farther."



Single choice poll

A. Ava



Single choice poll

B. Ben



Single choice poll

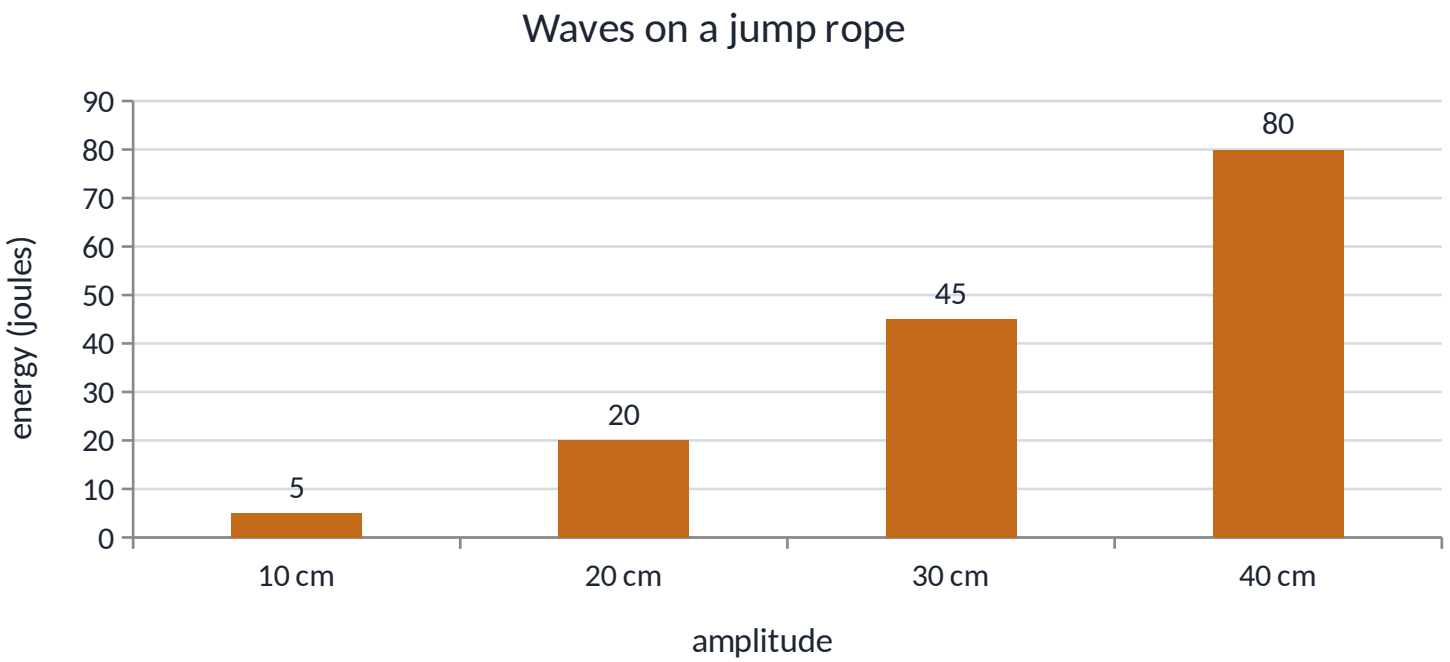
C. Cam



Single choice poll

D. Dee

Graph Poll



Read the graph.

Which statement does the graph support?

Single choice poll

A. As amplitude goes up, energy goes up.

Single choice poll

B. As amplitude goes up, energy goes down.

Single choice poll

C. Energy stays the same.

Single choice poll

D. Amplitude only changes frequency.

Key Ideas

1. Amplitude = how far the medium moves from its rest position to a crest or trough.
2. Bigger amplitude = more energy. Louder sounds have larger amplitudes.
3. Shake a rope or pluck a string harder: bigger amplitude, more energy.
4. Changing amplitude does NOT change wavelength, frequency or speed.

Re-rate your goals: 1 = Not yet, 2 = Getting there, 3 = I can teach it